

1 Approximate solutions of ODEs



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 14

Approximate solutions of ODEs

Thursday 8 October 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Approximating a solution of a first order ODE

Consider an IVP

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \quad (\heartsuit)$$

We will assume $f(x, y)$ and $f_y(x, y)$ are continuous, so we know there is a unique solution $y(x)$ near the initial point (x_0, y_0) .

It is often not possible to find a formula for $y(x)$. Nevertheless, we can still *approximate* the solution.

Let's approximate $y(x_0 + h)$ for small h . Approximate the derivative by a *difference quotient*:

$$f(x_0, y_0) = \frac{dy}{dx}(x_0) \approx \frac{y(x_0 + h) - y(x_0)}{h}.$$

Since $y(x_0) = y_0$, rearranging gives

$$y(x_0 + h) \approx y_0 + h f(x_0, y_0)$$

Geometrically, we jump along the tangent line at (x_0, y_0) .

Euler's method

Euler realized in 1768 that the **difference quotient approximation** yields a method for approximating the full solution $y(x)$ to (♥).

Choose a step size h and start from the initial point (x_0, y_0) .

Repeat:

$$\begin{aligned}x_{i+1} &= x_i + h, \\ y_{i+1} &= y_i + h f(x_i, y_i).\end{aligned}$$

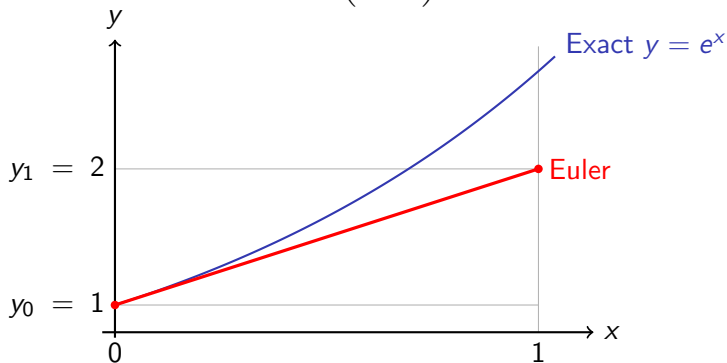
y_i is an **approximation** to $y(x_i)$.

At each step, we are evaluating the slope at the **current approximate point** (x_i, y_i) . Then we follow that slope for a distance h along the x axis.

Euler's method: tangent steps

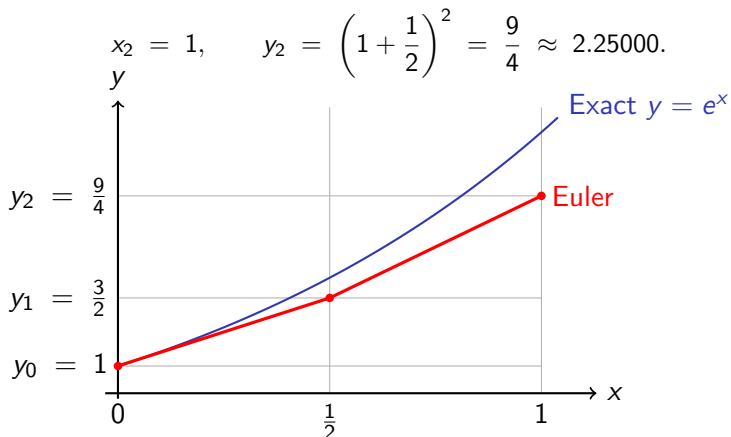
For $\frac{dy}{dx} = y$, $y(0) = 1$, the exact solution curve is $y = e^x$; the red line segments below show $n = 1$ Euler step, each of size $h = 1$.

$$x_1 = 1, \quad y_1 = \left(1 + \frac{1}{1}\right)^1 = 2 \approx 2.00000.$$



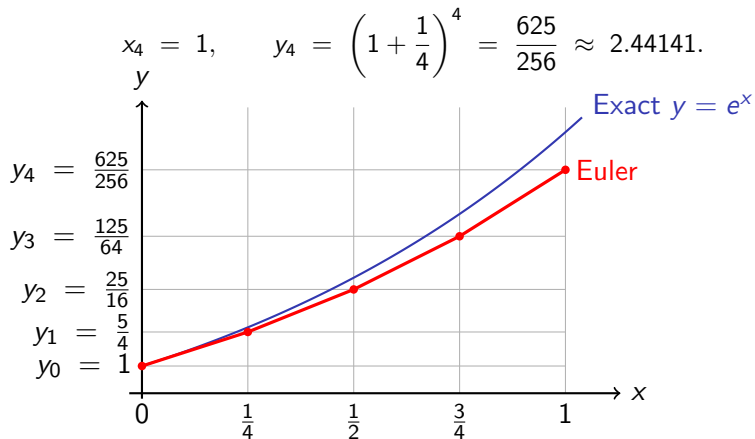
Euler's method: tangent steps

For $\frac{dy}{dx} = y$, $y(0) = 1$, the exact solution curve is $y = e^x$; the red line segments below show $n = 2$ Euler steps, each of size $h = \frac{1}{2}$.



Euler's method: tangent steps

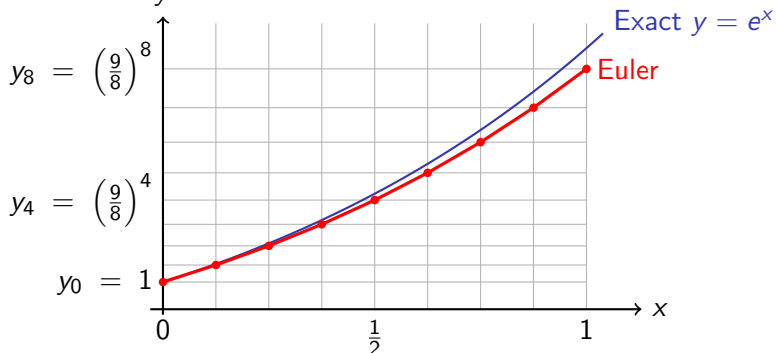
For $\frac{dy}{dx} = y$, $y(0) = 1$, the exact solution curve is $y = e^x$; the red line segments below show $n = 4$ Euler steps, each of size $h = \frac{1}{4}$.



Euler's method: tangent steps

For $\frac{dy}{dx} = y$, $y(0) = 1$, the exact solution curve is $y = e^x$; the red line segments below show $n = 8$ Euler steps, each of size $h = \frac{1}{8}$.

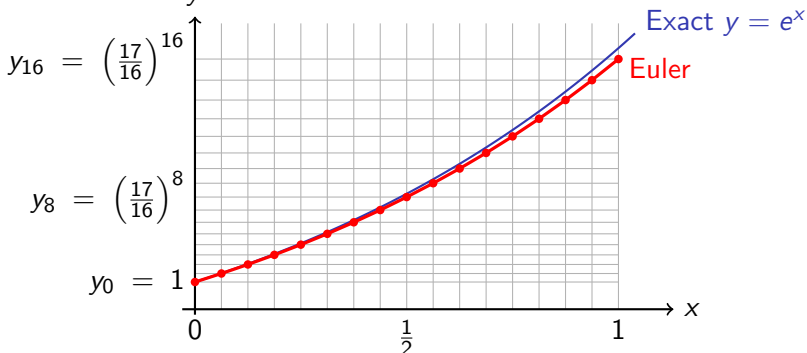
$$x_8 = 1, \quad y_8 = \left(1 + \frac{1}{8}\right)^8 = \frac{43046721}{16777216} \approx 2.56578.$$



Euler's method: tangent steps

For $\frac{dy}{dx} = y$, $y(0) = 1$, the exact solution curve is $y = e^x$; the red line segments below show $n = 16$ Euler steps, each of size $h = \frac{1}{16}$.

$$x_{16} = 1, \quad y_{16} = \left(1 + \frac{1}{16}\right)^{16} = \frac{486611918756668681}{18446744073709551616} \approx 2.63793.$$



Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Euler's line segments**
- .

Euler's method: the limit as $h \rightarrow 0$

For $\frac{dy}{dx} = y$ with $y(0) = 1$, the Euler update is

$$y_{i+1} = y_i + h f(x_i, y_i) = y_i + h y_i = (1 + h)y_i.$$

Starting from $y_0 = 1$, we therefore obtain

$$y_1 = 1 + h, \quad y_2 = (1 + h)^2, \quad \dots, \quad y_n = (1 + h)^n.$$

To reach a fixed $x > 0$ in n steps, choose $h = \frac{x}{n}$. As $n \rightarrow \infty$, $h \rightarrow 0$, and the limit is

$$y_n = \left(1 + \frac{x}{n}\right)^n \longrightarrow e^x = y(x)$$

- The previous slides showed this for $x = 1$.
- With smooth $f(x, y)$ in (♥), Euler approximations converge to the exact $y(x)$ as $h \rightarrow 0$ on any fixed closed interval where the solution exists, regardless of whether we can find an analytical expression for $y(x)$.
- In this example, we can also write down simple expressions for each Euler approximation and the exact solution.

Euler's method

Example (Two steps for $\frac{dy}{dx} = x + y$, $y(0) = 1$, $h = \frac{1}{2}$)

Here $f(x, y) = x + y$ and $(x_0, y_0) = (0, 1)$.

First step: $x_1 = x_0 + h = \frac{1}{2}$,

$$\text{so } y_1 = y_0 + h f(x_0, y_0) = 1 + \frac{1}{2}(0 + 1) = \frac{3}{2}.$$

This estimates $y(\frac{1}{2})$.

Second step: $x_2 = x_1 + h = 1$,

$$\begin{aligned}\text{so } y_2 &= y_1 + h f(x_1, y_1) \\ &= \frac{3}{2} + \frac{1}{2} \left(\frac{1}{2} + \frac{3}{2} \right) = \frac{5}{2}.\end{aligned}$$

Thus, Euler's two-step estimate is

$$y(1) \approx \frac{5}{2}$$

Practice problems: Euler's method

In [Trench, §3.1](#):

- Exercises 1–5: compute the first three Euler steps.
- Exercises 6–8: compare estimates for different step sizes.

For any exercise, keep a small table of (x_i, y_i) values so that each new slope is evaluated at the correct point.

Note: Computers are ideal for these repetitive calculations and can use tiny steps to obtain very accurate approximations. In practice, we use a computer rather than calculate a long table by hand. These exercises are just to help you understand how the method works.

Taylor's theorem and higher-order methods

Another derivation of Euler's method uses Taylor's theorem.

- If y'' is continuous, use $y'(x) = f(x, y(x))$ to obtain

$$y(x+h) = y(x) + hf(x, y(x)) + \frac{h^2}{2}y''(\xi),$$

where ξ is between x and $x+h$.

- Dropping the last term gives the [Euler approximation](#).
- If $|y''| \leq M$ on this step, Taylor's theorem also gives

$$|y(x+h) - [y(x) + hf(x, y(x))]| \leq \frac{Mh^2}{2}.$$

- This bounds the error in *one step starting from the exact point* $(x, y(x))$.

Taylor's theorem and higher-order methods

With sufficiently smooth f , keeping more Taylor terms gives higher-order methods. For example, the chain rule gives

$$y''(x) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = f_x(x, y(x)) + f_y(x, y(x)) f(x, y(x)),$$

so we can add $\frac{h^2}{2}(f_x + f_y f)$, evaluated at (x_i, y_i) , to the [Euler update](#) to obtain a [second-order method](#). Over a fixed interval, its accumulated error is bounded by a constant times h^2 .

For Euler, the error bounds involve h^2 for one step from an exact point, but h over a fixed interval. Keeping the next Taylor term improves these powers to h^3 and h^2 , respectively.

Many more clever techniques have been invented. Standard software solves ODEs accurately using sophisticated algorithms, often adjusting the step size automatically.

See [Trench, §3.1](#) for error analysis of Euler's method.

Truncation and roundoff errors

Numerical approximations have two sources of error:

- **Truncation error:** we truncate (cut short) Taylor's expansion by dropping terms. For Euler's method, the omitted term is

$$\frac{h^2}{2}y''(\xi).$$

Truncation error can occur even with exact arithmetic.

- **Roundoff error:** computers represent numbers with finite precision, so many numbers and arithmetic results must be rounded. For example, if we kept only four decimal places,

$$\frac{1}{3} \approx 0.3333.$$

This introduces an error into subsequent calculations.

Both kinds of error can accumulate over many steps. See [Trench, §3.1](#).